

What is the use of `search()` in Regular Expressions? Explain with an example.

Ans:

The `re.search()` function in Python's `re` (Regular Expression) module is used to **search for the first occurrence of a pattern** in a string. If the pattern is found, it returns a **match object**; otherwise, it returns **None**.

Syntax:

```
re.search(pattern, string)
```

Uses of `re.search()`:

- Finds the first match of a pattern in a string.
- Checks whether a specific pattern exists in the text.
- Returns the position and matched text if found.

Example:

```
import re

text = "Python is easy to learn."

result = re.search("easy", text)

if result:
    print("Match found:", result.group())
else:
    print("No match found")
```

Output:

```
Match found: easy
```

Explanation:

- `re.search("easy", text)` searches for the word **"easy"** in the given string.
- Since the word exists, it returns a match object.
- `result.group()` displays the matched text (`easy`).
- If the pattern is not found, `re.search()` returns `None`.